

Renewable-Energy PoC - Executive Summary

Chilean renewable-energy data platform on Databricks | updated 2026-07-21

Overview

A medallion (bronze -> silver -> gold) lakehouse on Databricks. It ingests real data from four independent sources across three domains, governed and monitored - all in dev, on branches; production is the project lead's.

Four data sources - all on real data

- Weather: NASA POWER (historical hourly) + DMC (real-time measured) + MinEnergia/DGF (modeled).
- Resource (generation): CEN/Coordinador - public by law, no cost.
- Conglomerate (country): OWID, open CSV.

Results (verified in dev)

- Medallion runs end-to-end for all three domains.
- Weather: 10-year historical hourly (876,720 rows), modeled as a star schema (dim + fact) with gold marts (KPI, daily, monthly) and an AI/BI dashboard (time series + KPIs). Atacama = highest irradiance.
- Resource: real per-plant generation (coordinado/real/reducciones).
- Conglomerate: Chile-vs-World renewables lead grew +19 -> +38 (2017-2024).

Governance & data quality (native)

- Data contracts per step + DLT expectations; data glossary; UC lineage + comments.
- Three Data-Quality Monitors on gold (profiling + drift) + a SQL alert.

Verify the medallion & jobs

- Resource pipeline: <https://dbc-54b27bae-2e91.cloud.databricks.com/pipelines/9bcfc32a-19f0-4327-b5bc-e8ee1cc4a3a4?o=7474645896934260>
- Conglomerate pipeline: <https://dbc-54b27bae-2e91.cloud.databricks.com/pipelines/b5d63282-a439-42ef-8e3b-cd707a9d5f58?o=7474645896934260>
- Weather NASA dashboard (series+KPIs): <https://dbc-54b27bae-2e91.cloud.databricks.com/sql/dashboardsv3/01f18771593d19568e23e322277007c3?o=7474645896934260>
- Weather NASA gold monthly: https://dbc-54b27bae-2e91.cloud.databricks.com/explore/data/workspace/dev_fuad_onate_renewable_gold_energy_chile/gold_weather_nasa_monthly?o=7474645896934260
- Quality monitor (Resource gold): <https://dbc-54b27bae-2e91.cloud.databricks.com/sql/dashboardsv3/01f1809adb1a199384b80e96c469a206?o=7474645896934260>
- Pull requests: https://github.com/oxiboy/poc_databricks_evaluateserve/pulls